

Federated Learning-based Misbehaviour Detection for the 5G-enabled Internet of Vehicles

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Abstract – The concept of federated learning (FL) is becoming increasingly popular as a method for training collaborative models without loss the sensitive information. The term has become ubiquitous due to the extensive development of autonomous vehicles. Vehicular Networks and the Internet of Vehicles (IoV) enable cooperative learning through federated learning. It is still necessary to address several technical challenges. In recent years, Federated Learning (FL) has attracted significant interest in various sectors, including smart cities and transportation systems. FL-enabled attack detection for IOVs are still in its infancy. However, to determine the main challenges of deployment in real-world scenarios, there needs to be research efforts from various areas. Performance metrics are used to evaluate the effectiveness of the proposed FL framework. According to experiments, the proposed FL approach detected attacks in IOV networks with a maximum accuracy of 99.72%. In addition to precision, recall, and F1 scores, 99.70%, 99.20%, and 99.26% were achieved. A comparison of the proposed model with the existing model shows that the proposed model is more accurate.

Keywords – Federated Learning, Internet of Vehicles (IOV), intelligent transportation system (ITS), 5G, and vehicular ad-hoc networks (VANETs)

I. INTRODUCTION

IOV-derived VANETs are an important research area in IoT. As sensor technologies grow rapidly, IOV generates large volumes of data. Transportation systems must meet the IOV requirements to be effective. Different companies in different countries have introduced intelligent transportation systems based on IOVs [1]–[4]. The most common IOV applications are traffic control systems, traffic flow monitoring, toll plazas and intelligent vehicle control.

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Cellular networks' fifth generation aims to improve reliability, throughput, delay, and connectivity while

improving service quality [5]. The Internet of Things (IoT), smart houses, Intelligent Transportation Systems (ITS), and health monitoring are some of the developing applications of fifth generation (5G) [6]. Physical things are associated with the Internet through the IOTs, an emerging revolution. As part of the IoT, the Internet of Vehicles (IOVs) connects vehicles to the Internet and allows them to communicate [7], [8]. IoV technology and Intelligent Transportation Systems (ITS) are evolutions towards vehicle-to-everything (V2X) tech. The V2X program aims to make roads safer, communication more reliable, and traffic flow more efficient [9]. Figure 1 shows four kinds of vehicles-to-vehicles (V2V), vehicles-to-infrastructure (V2I), vehicles-to-pedestrians (V2P), and vehicles-to-networks (V2N) communication—most ITS use machine learning techniques to provide comfort and safety to end users [10].

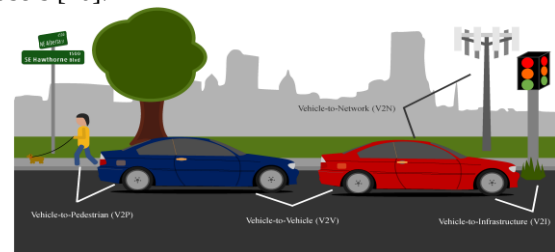


Figure 1: Internet of the vehicle and its communications.

The IoV has developed as a new standard in the field of ITS thanks to the advancement of IoT. As a result of the development of the traditional vehicle ad-hoc network (VANET) network, and road transport equipment, the IoV has evolved with the IoV [11]. In addition to driving assistance services, traffic notifications, speed advisory services, accident notifications, and emergency vehicle notifications, the IoV promises to redefine how users live and offer them better opportunities [12]. In the IOVs, a complex vehicular network connected to the Internet, sensors are installed on vehicles and roads to collect data. Vehicles, personal devices, and remote sensing units are connected to cloud networks through wireless communication networks based on various wireless access technologies. There are different preferences and requirements for each IoV and network device. Vehicles and other devices utilize smart cloud services based on their application requirements. Safety, traffic management, and commercial applications are different IoV applications. Smart cloud infrastructure processes ITS applications like safety, navigation, real-time traffic information management, and parking; the results are fed back to IOVs.

With the ubiquity of sensors in vehicular networks, it is feasible to capture more data and train ML models. Machine

Learning based models are generally applied to both vehicle management and traffic management [13]. The dynamic nature of the surroundings limits the current autonomous driving decisions as the training is carried out offline. FL can rescue such situations by online training vehicles from different geographical locations, facilitating accurate labelling of the features. Similarly, a large amount of data is required for traffic flow prediction techniques. Still, most data is divided among various organizations and cannot be exchanged to protect privacy [14]. To address such situations also, we can deploy FL methods.

1. In the paper proposed Federated Learning-based Misbehaviour detection for the 5G-enabled Internet of Vehicles (IOVs) using federated distillation (FD) training scheme with FL baselines.
2. In this paper, Four network attack detection datasets are evaluated, namely ISCXIDS2012 [15], CIC-110 IDS2017 [16], CSE-CICIDS2018 [17], and Car-hacking [18].
3. As a benchmark, other attack detections FL baselines are compared to evaluate the effectiveness of this method.

The rest of the paper is structured as follows: Section 2 explains the literature review based on the latest attack detection techniques using Federated Learning in the field of IOV. Section 3 presents the proposed Federated learning-based framework with complete mathematics. Section 4 presents the experimental setup, datasets, result discussion, and comparative analysis with the FL baselines. Section 5 presents the conclusion.

II. LITERATURE REVIEW

A rapidly growing market, especially the Internet of Things (IoT), is moving toward 6G networks [19], [20]. This raises security concerns significantly [21]. The Internet of Things and its Values reshape networking and communication [22]. The Internet of Things/Internet of Vehicles can be defined differently [23], usually encompassing physical objects like sensors and vehicles [24]. In addition, the meaning has broadened from local to global, encompassing both local area networks (LANs) and next-generation 6G networks. The reasons for this are that 5G technology cannot handle the IoT/IoV applications requirements of the near future [25]. IoT/IoV networks are experiencing much greater traffic as more devices are added. In one sense, IoT/IoV networks enrich our everyday lives and industrial equipment in the other. The number of malware threats is also increasing across all layers and phases of the life cycle of the network [26], [27]. Due to the heterogeneity and dynamic nature of IoT and IoV, security has become one of the most important research areas [22]. 6G networks are expected to support many heterogeneous devices and infrastructures, surpassing 5G networks in many ways [28]. Table 1 shows some common vehicular cyber-attacks.

Table 1: The description of cyber-attacks in the IOV network

Attack	Description
DoS	In a DoS attack, malicious actors attempt to disable a device's normal operation to

	prevent its users from using it.
Jamming	An example of a jamming attack is when a malicious node interferes with a network to disrupt legitimate communication.
Spoofing	Spoofing is used in cyber security to describe the act of impersonating another entity to gain trust, gain access to systems, steal data, steal money, or distributes malware.
Acoustic	Computers and smart devices are vulnerable to acoustic attacks.
Eavesdropping	A technique of passively listening to network conversations is called eavesdropping, in which an attacker obtains private information such as node identifiers, routing changes, and application-sensitive information.
Blackhole	Routers that delete all messages supposed to be forwarded are said to be committing a blackhole attack. A router can be set incorrectly to give a zero-cost route to every destination on the Internet occasionally.
Cloaking	Users' credentials are often obtained through social engineering attacks on malicious websites impersonating well-known businesses. It is common practice for certain websites to hide hazardous information from search engine crawlers while displaying it to users/client browsers, known as cloaking.
Replay	Hackers typically perform replay attacks by intercepting and fraudulently delaying or resending data from a secure network connection to trick the receiver into doing what they want.

Federated Learning was introduced by Google in 2017 [29], which enables machine learning models to be trained with data collected across multiple mobile devices while preserving user privacy. Researchers have been interested in federated learning since its inception, focusing on data privacy, communication efficiency, and statistical heterogeneity [30]. To address the challenge of data heterogeneity in the failure detection of intelligent interconnections of things, [31] presents a blockchain-based federated learning system platform. [32] Provide possible solutions for wireless end-user equipment security issues present in Federated Learning. A federated reinforcement learning (FRL), which combines edge computing and distributed deep learning, has also been proposed by [33] to improve computational efficiency for urban environment sensing. Table 2 presented the comparison of previous survey articles on the Internet of Vehicles (IoVs). According to the authors [34], on-device decentralized data can be used to detect federated IoT attacks at the edge layer. Instead of taking the data to an offsite computation, this software solution uses Federated Learning (FL) [29] as its core. Because FL does not require data to be transported from the network on which the devices are located, it allows for faster machine-learning computation on local devices. Data

security concerns are best addressed this way. By minimizing sensitive data movement and providing quick-acting services that react to FL results, the security of IoT data will increase in response [35], [36].

The fourth industrial revolution and modern communication technology have resulted in increased use of digital twins (DT), federated learning (FL), and the Internet of Things

(IIoT). It is, however, challenging to deploy DT and FL for IoV. IoT, IoV, and Internet of Drones (IoD) applications of DT and FL are presented [37]. As a result, BC-enabled FL solutions can be developed to address different IoV scenarios and data privacy issues. For evaluation, real-time data are collected regarding denial-of-service (DoS) and shellcode attacks, as well as normal data.

Table 2: Comparison of previous survey articles on the Internet of Vehicles (IoVs)

Author & Year	Objectives	Keywords	FL	IoVs + FL
[38]	Secure federated-learning framework	Security & privacy	Yes	No
[35]	Present deep research on FL enabling, protocol, Technologies and Applications	Security & privacy, collaborative AI	Yes	No
[39]	Mobile Edge Computing (MEC) is used to bring intelligence closer to the edge	cloud computing, data privacy, resource allocation	Yes	Not
[40]	FL presents a solution to privacy & security issues	smart cities sensing, IOTs, security, privacy	Yes	No
[36]	FL's security and privacy aspects integrated with federated Artificial Intelligence (AI)	Security & privacy, AI, ML and Distributed Learning	Yes	No
[41]	Presents a deep survey on privacy preservation with centralized FL	FL, Data protection regulation, Personal Data, Privacy, Privacy preservation	Yes	No
[42]	Present an integration of IIoT and FL for achieving data privacy preservation.	ML, deep learning (DL), blockchain techniques with secure IIoT	Yes	No
[43]	FL with five characteristics: privacy mechanism, data partitioning, ML model, heterogeneity system and communication architecture	FL, Privacy protection, ML	Yes	No
[44]	Presents FL with two-level Multi-access Edge Computing (MEC) assisted vehicular network	AI, MEC, IOV, 5G, ITS	Yes	Yes
[45]	Proposed a Blockchain-based Privacy-preserving Federated Learning scheme (BPFL)	FL, Homomorphic encryption, IOVs, Privacy-preservation	Yes	Yes
[46]	Proposed an optimal resource allocation approach based on FL.	DL, IOVs, scheduling, resource allocation	Yes	Yes
[47]	Analyze Blockchain-enabled FL frameworks for IoV applications and implementations.	IoVs, Blockchain, FL, ML	Yes	Yes
[48]	Proposed a robust hierarchical FL system named RoHFL. Hierarchical FL applied against poisoning attacks.	Hierarchical FL, poisoning attacks, robustness, IOV	Yes	Yes

A. Federated Learning in IoVs

FL for developing IoV has extensively been studied in the literature. This research works on different application areas, including Unmanned Aerial Vehicles (UAV) [49], [50], autonomous Driving [51], [52], electric vehicle Networks [53], [54], Vehicle Selection and Resource Optimization, traffic flow Prediction [14], distributed map management, Content Caching, vehicle positioning and parking space estimation [55], [56] etc. as summarized in Figure 2.

An overview of Blockchain (BC) Enabled Federated Learning frameworks for IoVs is presented [47]. Moreover, BC-Enabled FL frameworks for IoVs are examined regarding probable issues, challenges, solutions, and future research directions.

The author [57] presents an overview of current ML, DL, and Explainable Artificial intelligence (XAI) techniques [58]. The findings provide helpful advice to researchers considering integrating ML and DL models into the IDS network. Since vehicles are reluctant to upload local data directly to the RSU due to privacy and security concerns, federated learning (FL) can be used in vehicular edge computing (VEC) for ML tasks,

where vehicles only have to upload their local model hyperparameters rather than transferring their local data to a nearby RSU, making it a promising technology [59]. Vehicle-to-Everything (V2X) nodes need to collaborate when developing ML attack detection models for 5G and Beyond (5GB) vehicular networks [60]. Author [61] propose a new Internet of Vehicles system that utilizes three Blockchain to facilitate the detection and sharing of real-time information.

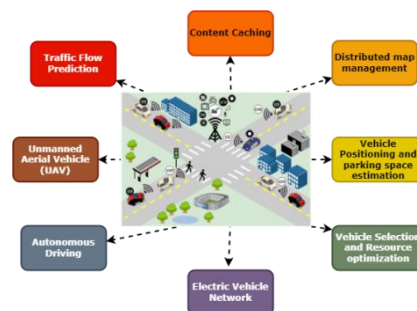


Figure 2: Application of Federated Learning (FL) in IOVs.

III. PROPOSED FEDERATED LEARNING-BASED FRAMEWORK

These concerns have been addressed by Federated Learning (FL) [29], [62], [63]. An FL device will not share any data instance with anyone else in the system; however, a centralized server coordinating FL will share the Machine Learning (ML) model. The coordinator combines the local models (local models) to build an aggregated model (global models) based on the aggregated data from each device. FL involves sharing ML models, thus protecting confidential data. The basic idea is not to disclose data that may contain confidential information. It faces various performance challenges compared to centralized learning [63]. Performance degradation is believed to be caused by the limited data each device generates, which limits its effectiveness. Devices participating in FL environments typically encounter different data types; this class imbalance can severely affect performance. Additionally, heterogeneous devices may observe different events when participating in the event, further degrading performance when devices of different types participate. Many FL settings face these challenges due to the large number of end devices.

FL can be used in vehicular IOV networks for several reasons, including (i) ML system settings can be configured locally instead of exchanging a large amount of training data; (ii) Locally calibrating ML model parameters can greatly reduce transmission delay; (iii) As only the local variables of a learning model are sent, FL can help protect data privacy. In FL, edge devices create a learning model by sending only local information to learned designs to a global aggregation server [64]. FL for vehicular IOV networks is illustrated in Figure 3 and systematic diagram of the proposed FD-Scheme for attack detection in the IOV Network is mentioned in Figure 4.

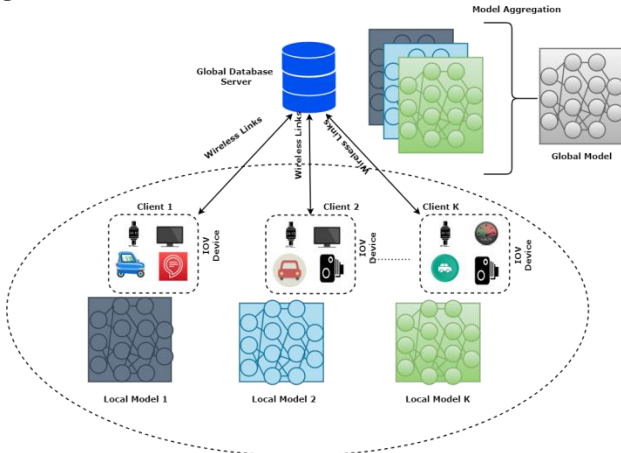


Figure 3: A general diagram of FL for Vehicular IoV devices.

A. FL Scheme

A FL approach is designed in four steps. A DL model is first trained with private data sets by IoV or vehicular devices participating in FL. In the second step, every client uploads the model's parameters to the central server. Servers calculate the average value of the constraints obtained from every client by adding their parameters. In the end, each client receives the

new parameters and continues the loop until convergent training occurs.

In particular, consider that K clients contribute in FL $k = 1, 2, 3, \dots, K$, and each has a privately labelled dataset. $D^k = \{(x_i^k, y_i^k) | i = 1, 2, \dots, n\}$, where x_i^k , y_i^k , and n represent the vectorized features of samples, respectively, and where D^k represents the number of instances in the dataset—a classification task involving L -classes. Y_i^k is a one-hot vector, i.e., $y_i^k = y_{i,0}^k, y_{i,1}^k, \dots, y_{i,L-1}^k$. $\{x_i^k | i = 1, 2, \dots, n\}$ and $\{y_i^k | i = 1, 2, \dots, n\}$ are concatenated respectively into matrices X^k and Y^k .

DL models of the K client and server are initialized prior to the training of the first communication round, i.e., $w^k \leftarrow w^s$, where w^s and w^k represent the initial parameters of the server DL model and the initial parameters of the client DL model, respectively.

With the private data set provided by client k , client k trains its DL model in the training step of the first communication round. Once the model parameters w^k have been uploaded to the server. Each $k = 1, 2, 3, \dots, K$ client uploads them. To generate new parameters, the server aggregates the parameters as follows: w^s as mention in Eqn. (1)

$$w^s = \sum_{k=1}^K \frac{N^k}{N} w^k \quad (1)$$

There are N private samples in total, i.e., $N = \sum_{k=1}^K N^k$. Second, new DL parameters will be transmitted to every client, i.e., $w^k \leftarrow w^s$. This process will be repeated for various communication rounds.

B. FD Scheme

The FD scheme eliminates client responsibility for uploading parameters; clients only need to upload averaged predictions from each label, significantly reducing the number of communications throughout training [65]. As a result of the labels assigned to the private dataset maintained by client K , the dataset can be further separated into L subsets, i.e., $D^k = D^{k,0} \cup D^{k,1} \cup \dots \cup D^{k,L-1}$. $y_i^{k,l} = [y_i^{k,l}, y_{i,1}^{k,l}, \dots, y_{i,L-1}^{k,l}]^T$, represents the label of each sample $y_i^{k,l} = 1$ and others are 0 for the subset $D^{k,l} = \{(x_i^{k,l}, y_i^{k,l}) | i = 1, 2, \dots, n\}$. The FD method consists of six steps.

Each client K first trains its local model. w^k using the same dataset D^k , as FL. Every client then measures a local-average logit vector (LALV). Detailed information can be found in Eqn.(2-3) below:

$$\hat{y}_i^{k,l} = F(x_i^{k,l} | w^k) \quad (2)$$

$$\bar{y}^{k,l} = \frac{1}{N^{k,l}} \sum_{i=1}^{N^{k,l}} \hat{y}_i^{k,l} \quad (3)$$

The label on sample l may not be available to client k , i.e., $D^{k,l} = \emptyset$. When such a situation occurs, label l has a local-average logit vector of zero, i.e., $\bar{y}^{k,l} = 0$. Once the LALV $\{\bar{y}^{k,0}, \bar{y}^{k,1}, \dots, \bar{y}^{k,L-1}\}$ have been uploaded by each client, and the server receives them. In the next step, each client receives the global-average logit vectors (GALV).

These LALV are computed by each client according to Eqn. (4).

$$\bar{y}^{k,l} = \frac{1}{N^l - 1} (N^l \times \bar{y}^{s,l} - \bar{y}^{k,l}) \quad (4)$$

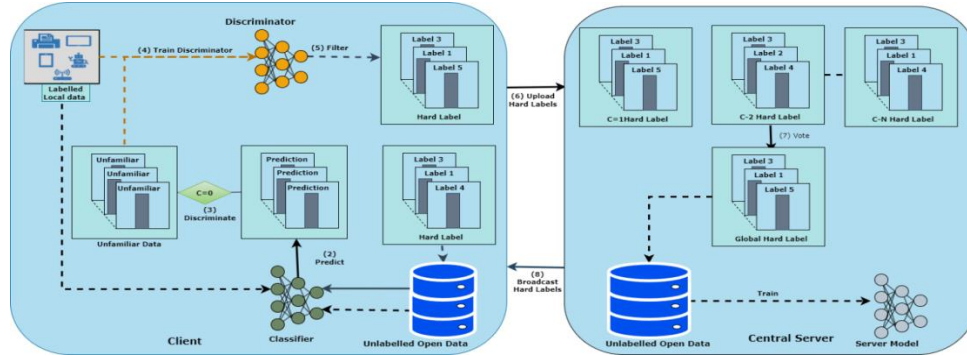


Figure 4: The systematic diagram of the proposed FD-Scheme for attack detection in the IOV Network.

IV. EXPERIMENTAL RESULTS

As part of the validation of this framework, different experiments were conducted to obtain results. The paper's first part compares supervised and unsupervised solutions for their performance when detecting malware.

A. Datasets and Pre-Processing

Two types of datasets are used in our experiments. In the first category, communications are conducted between the IOVs and external entities, such as infrastructure (vehicle-to-everything), and in the second category, communications are conducted within the vehicle. The datasets used in this study are described below:

- ISCXIDS2012 [15]: This distribution is heavily skewed and has binary labels "normal" and "attack".
- CIC-IDS2017 [16]: It is proposed to improve ISCXIDS2012 by including HTTPS, which is 70% of network traffic. ISCXIDS2012 is a multi-class dataset with a better class distribution than the improved dataset.
- CSE-CIC-IDS2018 [17]: Besides being a multi-class dataset, this is the third release in the series. In CIC-IDS2017 and CSE-CIC-IDS2018, one type of network traffic is categorized as "Benign", and the others are classified as attack types and subtypes.
- Car-Hacking [18]: A dataset of in-vehicle network traffic that includes malicious traffic from an actual vehicle's OBD-II port and traffic from its genuine network traffic. Among the attack categories in the CAR-HACKING dataset are four: spoofing of drive gears, spoofing of RPM gauges, and DoS attacks.

There is an uneven distribution of classes in the datasets, as shown in Table 3. 97.20 % of samples in the ISCXIDS2012 dataset are categorized as "normal", whereas only 2.8% are categorized as "attack". There are six categories of samples in the CIC-IDS2017 dataset, five in the CSE-CIC-IDS2018 dataset, and 80% and 83% of samples are considered "normal" according to the CIC-IDS2017 dataset. There are four different attack categories in the CARHACKING dataset,

Every sample $x_i^{k,l}$ Whose ground truth label l equals $\bar{y}^{k,l}$ has its local-average logit vector $\hat{y}_i^{k,l}$. The next step is for client k to train its model using local-average logit vectors and ground truth labels.

with 86.6% classified as "normal". An accurate and predefined class label is usually the target variable.

Table 3: The description of cyber-attacks in the IOV network

Dataset	No. of Instances	Attack Class	Label Type
ISCXIDS2012	2,450,324	2.8%	Binary
CIC-IDS2017	2,830,743	19.7%	Multi-class
CSE-CIC-IDS2018	19,233,002	17.0%	Multi-class
CAR-HACKING	17,357,681	13.4%	Multi-class

To ensure consistency, the range of data is normalized or scaled by converting it into a fixed scale. By using this technique, the performance of the network will be improved. Data input must be normalized using the following Eqn. (5) using a matrix format for deep learning:

$$d' = \frac{d - \min(d)}{\max(d) - \min(d)} \quad (5)$$

B. Performance Metrics

True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) are the four classifications of results used for result evaluation. An example dataset contains data packets that are normal and abnormal. A packet's classification is either true or false, depending on whether it is correct or incorrect. The model successfully detected an attack (TP means abnormal packets were detected), and FP means no abnormal packets were detected. Still, normal packets were classified as abnormal. TN means normal packets were detected as normal packets, while FN means some abnormal packets were misclassified as normal packets. This research was concluded using well-known metrics such as accuracy (A), precision (P), recall (R), and F1-score (F) [66]. The accuracy is calculated by the Eqn. (6).

$$Accuracy(A) = \frac{TP + TN}{(TP + TN + FP + FN)} \quad (6)$$

1) Recall (R)

Machine learning and pattern recognition rely on recall as a performance metric. Model sensitivity is measured with this tool. As a proportion of the total number of packets, the recall represents how many anomalous packets were detected. In Eqn. (7), we can calculate R.

$$Recall(R) = \frac{TP}{TP + FN} \quad (7)$$

2) Precision (P)

In terms of precision, it refers to the number of original packets detected among those detected. Measuring false alarms directly correlates with precision metrics. As false alarms are detected over the correct detection of anomalies, false positives (FP) are measured. False alarm refers to a system that misleads the user by detecting threats when none exist. It is possible to obtain P in the same way as in Eqn. (8).

$$Precision(P) = \frac{TP}{(TP + FP)} \quad (8)$$

3) F1-Score

In addition to F1-score, there are other performance metrics. F1-score can be calculated as the weighted average of recall and precision as mentioned in Eqn. (9).

$$F = \frac{2 * (Precision * Recall)}{(Precision + Recall)} \quad (9)$$

C. Results Discussion

IOVs selection strategies and misbehaviour detection accuracy are compared using baselines. PyTorch 1.8.1 and a Graphics Processing Unit (GPU) were used to develop the application on the Google Collaborator platform using Python 3.7. The complete software and hardware details are mentioned in table 4. IOVs are trained locally with the simulated attack datasets in all frameworks.

Figure 5 gives the average attack detection accuracy using the proposed FD-Scheme approach with FL baselines for the CIC-IDS2017, ISCXIDS2012, Car-Hacking and CSE-CIC-IDS2018 datasets. As can be seen from Figure 5, the accuracy for the FD-Scheme approach for all experiments came at 98% to 99%, where the average accuracy of FL baselines such as FL-Scheme and FL-Global reached 95% and 96%, respectively.

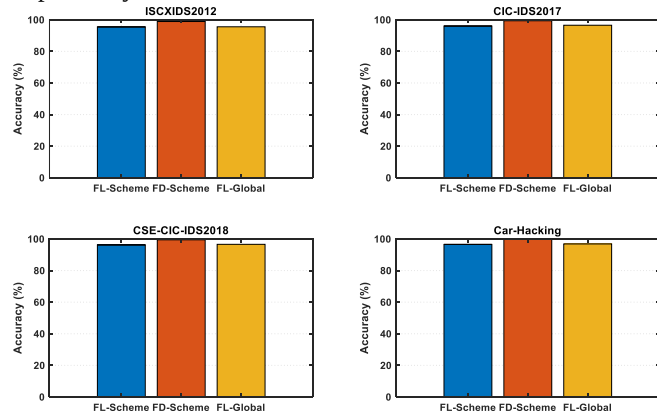


Figure 5: Accuracy (%) performance analysis versus proposed model FD-Scheme with FL baseline.

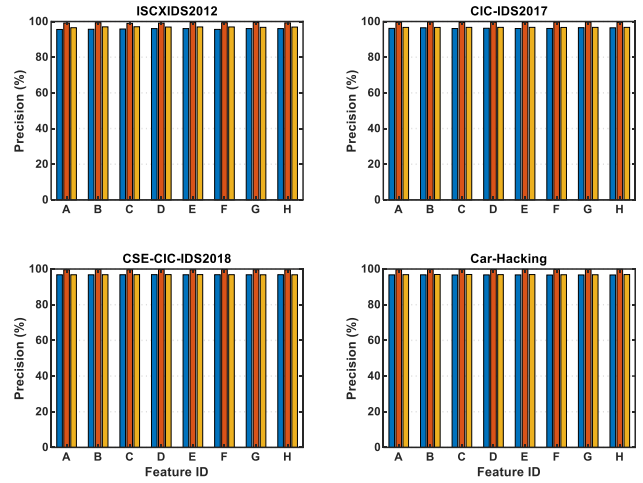


Figure 6: Precision (%) performance analysis versus randomly selected Feature.

Table 4: shows the complete software and hardware requirements.

Hardware	CPU	Intel® Core™ i7-7800xCPU @3.5GHz 32GB
	GPU	NVIDIA GeForce RTX 2080Ti 12GB
Software	OS	Linux Ubuntu 18.04.3 LTS
	Programming Language	Python 3.7
	FL Framework	PySyft, OpenFL

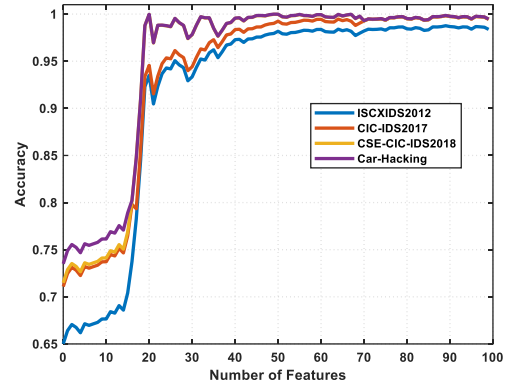


Figure 7: Accuracy versus the number of features.

For FD-Scheme, we evaluated its local performance on each Feature ID compared with the FL-Scheme and FL-Global baseline. As the results are shown in Figure 6, in the evaluation of CIC-IDS2017, ISCXIDS2012, Car-Hacking and CSE-CIC-IDS2018 dataset, except for each feature ID, A, FD-Scheme performs better or nearly equal to the FL-Global. The numbers above the bar are the relative difference between FD-Scheme and FL-Scheme settings. For example, feature ID A and H achieve a 99.4% to 99.8% accuracy increase from FD-Scheme for CIC-IDS2017, ISCXIDS2012, Car-Hacking and CSE-CIC-IDS2018 datasets, respectively. The FD-Scheme shows higher and more effective results on the Car-Hacking

dataset versus feature ID. Here randomly, 10 features are considered to calculate the performance.

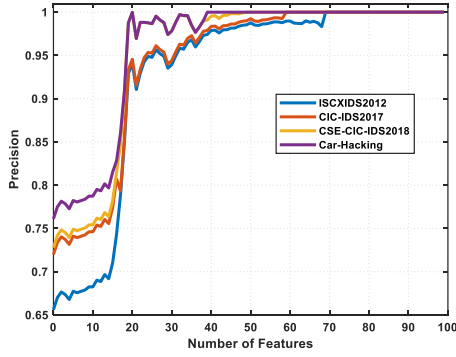


Figure 8: Precision versus the number of features.

To determine the minimum number of features required to detect IOV attackers, we vary the features and evaluate the accuracy, precision, recall and F1-Score performance. Figure 7 illustrates the accuracy evolution versus the number of features for all four datasets, ISCXIDS2012, CIC-IDS2017, CSE-CIC-IDS2018, and Car-Hacking, with the FD-Scheme approach. The results confirm that the proposed FD-Scheme performs better for the Car-Hacking dataset than the number of features. CIC-IDS2017, CSE-CIC-IDS2018, Car-Hacking dataset achieved 99% accuracy. Figure 8 shows the 100 % precision value for CIC-IDS2017, ISCXIDS2012, Car-Hacking and CSE-CIC-IDS2018 datasets, which is sufficient to obtain almost 69 features, 59 features, 50 features and 39 features, respectively.

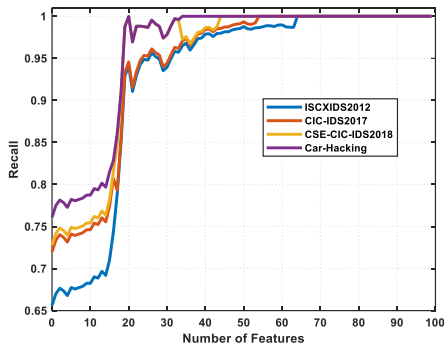


Figure 9: Recall versus the number of features.

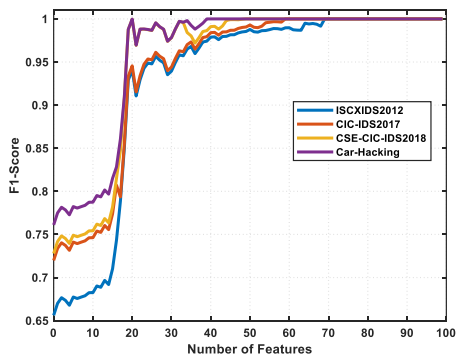


Figure 10: F1-Score with respect to the number of features.

Figure 9-10 shows that in the recall and F1-Score, with respect to the number of features, only 50 features are needed to detect the attack (100 % of each of the evaluation metrics) for all the datasets.

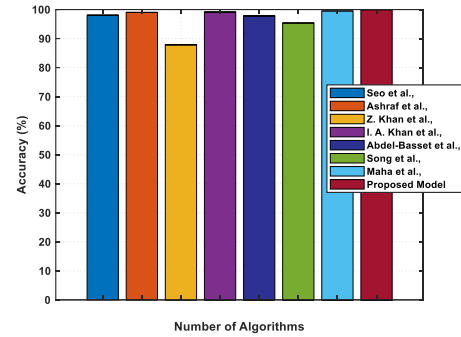


Figure 11: Accuracy (%) versus the number of existing algorithms.

To validate the performance of the proposed scheme, we compare the results with existing techniques such as GIDS [18], deep learning-based IDS [67], FED-IDS [68], DL-centered Bidirectional Long Short Term Memory (LSTM) [69], In-Vehicle Network Security [70], LSTM neural-network-based model [71] and FL-based attack detection framework for Vehicular Sensor Networks (VSN) [72]. The proposed model performed better than the methods presented on the Car Hacking dataset. According to Figure 11, the proposed model achieves higher accuracy, as in Table 5.

Table 5: Accuracy (%) versus the number of the existing algorithm of the IOV network

Author & Year	Accuracy (%)
[18]	98.0
[71]	99.11
[67]	99.0
[68]	97.82
[69]	97.82
[70]	95.37
[72]	99.52
Proposed Model	99.89

V. CONCLUSION

The work proposes a privacy-preserving IoV malware detection framework that uses FL for both supervised and unsupervised training and evaluation. Different Internet of Vehicles (IoVs) problems are being addressed through these new efforts. A 5G-enabled Intelligent Transportation System (ITS) can use new technologies. Due to its heterogeneity and divisibility in terms of devices and malware samples, the CIC-IDS2017, ISCXIDS2012, Car-Hacking and CSE-CIC-IDS2018 datasets have been used to demonstrate its feasibility. IOV attack detection is implemented using the FD-Scheme framework based on the extracted, pre-processed simulated attack dataset. Comparing the FD-Scheme to baseline frameworks based on FL, it is also shown that FD-Scheme achieves greater detection accuracy with fewer global aggregations. The proposed FD-Scheme approach detected

attacks in IOV networks with maximum accuracy, precision, recall, and F1 scores; 99.72%, 99.70%, 99.20%, and 99.26% were achieved. In the future, it is planned to create scalable solutions that can generate heterogeneous scenarios. The generated heterogeneous real-time data are applied in the proposed model.

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